Lab Activity #8 – Zero-Offset VSP Processing

We now seek to investigate the world of *Vertical Seismic Profiling* by processing a Zero-offset VSP data using SeismicUnix (su).

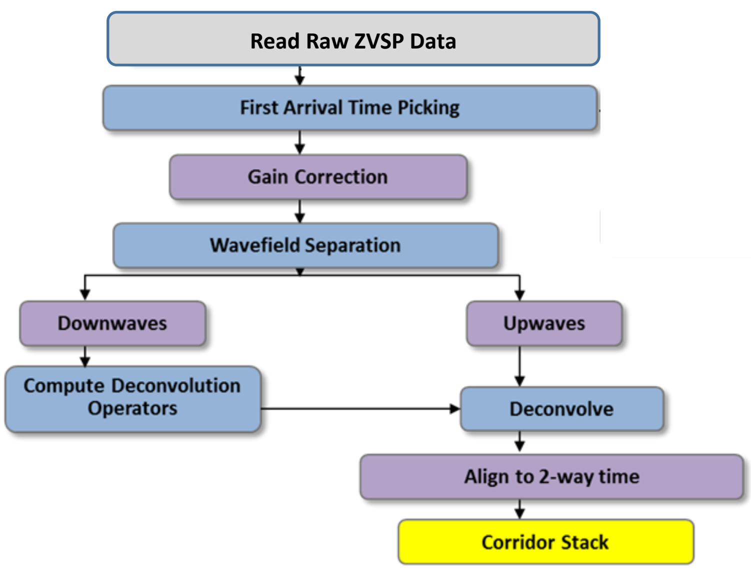
VSP Overview

Vertical seismic profiling (VSP) is a kind seismic survey where the seismic receivers array is deployed inside a wellbore and the source is placed on surface at some offset from the wellbore, depend on the desired survey and configuration. In the case of Zero-offset VSP, the source is placed at the nearest allowed position to the well bore. This kind of survey is generally part of well-logging, combined with other logging such as radioactive logging (gamma ray), density, and neutron, sonic logging etc.

The main purpose of running Zero-Offset VSP is to obtain the time-depth relationship at the well location as well as obtaining the corridor stack to be used for well-to-seismic calibration and performing well-to-seismic tie, among other applications such as Q estimation.

In this lab we will perform a basic zero-offset VSP processing workflow using a zero-offset dataset. You will be provided with a sequence of su scripts to run after you modify the input parameters as you see most fitting. In addition, several utility scripts are provided to help you reach to your desired input parameters as will be described below.

A basic ZVSP workflow is as follow



Please access the ZVSP folder (directory) and first, copy the ZVSP dataset: **Z.su**

In addition, copy the following files (su scripts) to your local working directory:

```
0_1_utl_ImportSEGY.sh
0_2_utl_Display_ZVSP_RAW.sh
0_3_utl_CheckBPFandAmpRecovery.sh
0_4_utl_FrequencyAnalysis.sh
0_5_utl_CheckAutoCorrelation.sh
1_FBAutoPick.sh
2_Preprocessing.sh
3_Separation.sh
4_ApplyPEFDecon.sh
5_CorridorStack.sh
```

The top five (5) su scripts are provided as utility scripts to help you observing the ZVSP datasets and decide on some of the input parameters you will use in your actual ZVSP processing workflow. The bottom five (5) su scripts represent the basic processing workflow numbered with the proper sequence which you will be asked to follow in this lab.

To learn more about the scripts feel free to open the files and browse through them. But first please learn more about your ZVSP dataset by running the following su command:
surange < Z.su &

This command will prompt you with the header values for this ZVSP dataset. Please read carefully.

Before you start running the processing sequence, start by displaying the ZVSP dataset by opening the running the following script: **0_2_utl_Display_ZVSP_RAW.sh**

Identify the down-going and up-going parts of the ZVSP.

Also analyze the frequency and wavenumber content of the data by opening and running **0_4_utl_FrequencyAnalysis.sh**. This script will also help you deciding on the cuttoff frequencies when you need to run a Band-Pass-Filter (BPF) later in the processing sequence.

In order to investigate and decide on the best parameters you can use for the amplitude recovery (gain) and BPF, use the following utility su script: **0_3_utl_CheckBPFAmpRecovery.sh**. Please open the script to modify the set of parameters and run for each parameter set until you are comfortable with the final parameters.

Now you are ready to start the processing sequence by running the script 1**_ FBAutoPick.sh** (Automatic First Break picking program by su). Feel free to open the script to learn how the script is written.

The next step is to perform preprocessing (cleanup) processes by applying a proper BPF and amplitude recovery/balancing by using the following script: **2_Preprocessing.sh**.
Note that

1. You need to run the **0_3_** and **0_4_** scripts to identify the best parameters to use in the script **2_Preprocessing.sh**.
2. You need to edit the script **2_Preprocessing.sh** to enter your final set of parameters of choice.

Now you are ready to run the separation process to separate the downgoing wave from the upgoing wave by running the following script: **3_Separation.sh**

Please open the script 3**_Separation.sh** to edit any required input parameters and learn more how the script is written. Two su files will be resulting from this process: the downgoing and upgoing waves sections. Please observe the two sections.

Now you are ready to use the downgoing wave to determine the best PEF parameters to use on the upgoing wave by running the utility script **0_5_CheckAutoCorrelation.sh**. Open the script to edit any needed input parameters. Determine the PEF input parameters of choice to be used in the next step.

Now, you are ready to open the script 4_ApplyPEFDecon.sh to run your desired decon and remove the upgoing multiples. Before you run the script, please open the file and edit the needed input parameters per your final choices but be consistent with what you have used in the previous steps.

Observe the differences between the upgoing waves section before and after the PEF Decon filter.

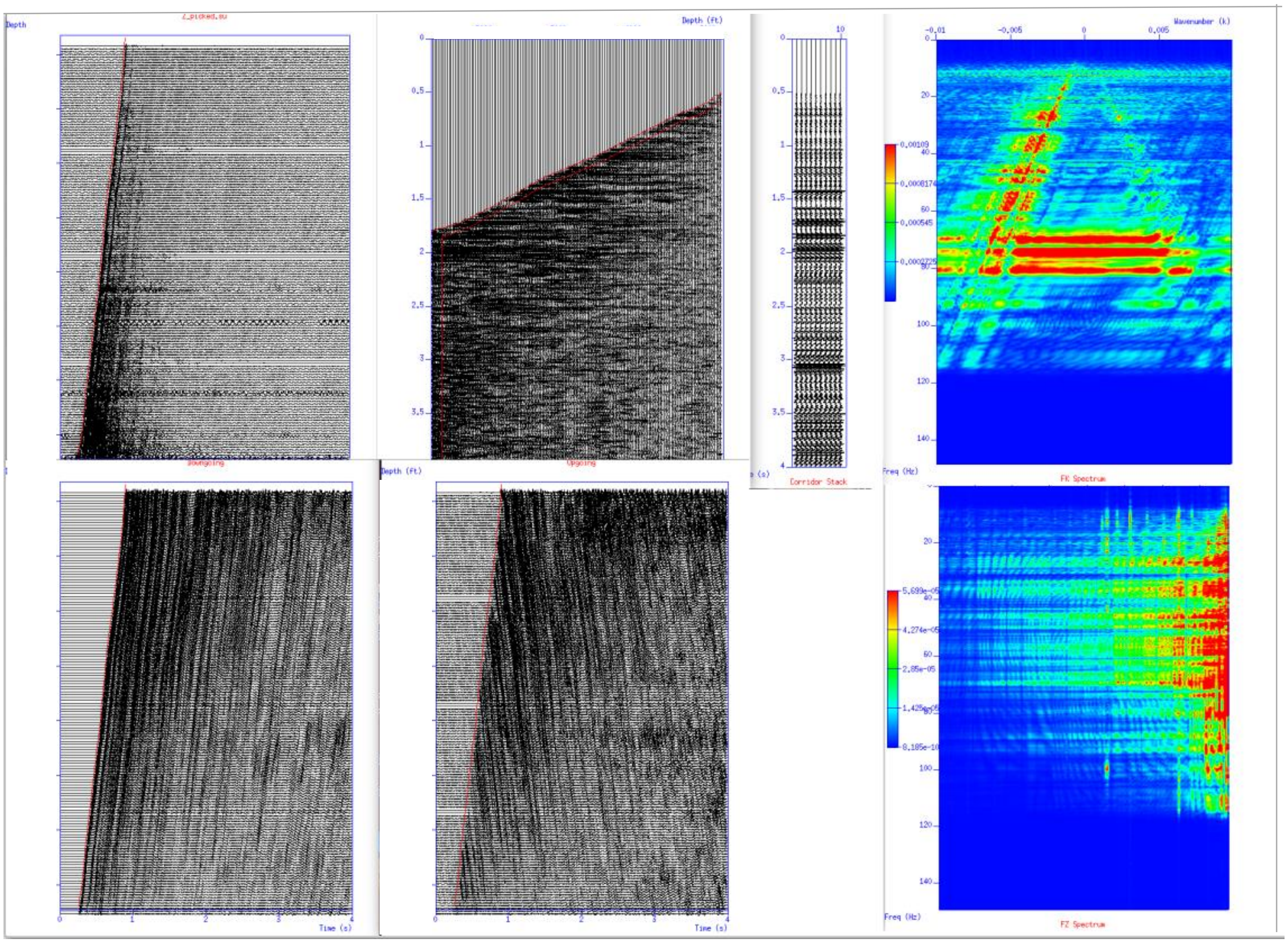
Now you are ready to run your final step in generating the corridor stack by running the final script: 5**_CorridorStack.sh**.

Open the script to Edit the input parameters per your final choices then you run the script. This script will result in a display of the flattened upgoing wave with the corridor stack window highlighted, in addition to the actual corridor stack trace displayed, repeated ten (10) times.

In the following page you will find a sample of some of the displays which will be generated by running the mentioned scripts above.

Estimate the Velocity Profile

The Corridor Stack is the final product of the zero offset VSP processing. The other product of the VSP processing is the velocity profile. This includes interval, average and rms velocities; all calculated from the first break picks and receiver depths. Use the first break picks from the ZVSP seismic data to estimate the velocities as a function of depth and time. You can develop an Excel spreadsheet to perform your calculations and plot the velocities.



Deliverable:

Please submit a 1-3 page summary report of your finings of the above steps. Support your summary with displays as you see most fitting.